ExoScan

A Physics-Validated, Uncertainty-Aware Pipeline for Detecting and Classifying Exoplanet Transit Signals in Noisy TESS Light Curves

Bharatiya Antariksh Hackathon 2026 — Problem Statement 7 — Methodology Report

1. Problem Summary and Approach Overview

Transit photometry detects exoplanets by measuring the small periodic dimming of a star as a planet crosses its disk. In crowded fields, this signal is confounded by instrumental systematics, stellar blending, starspot variability, and eclipsing binaries — all of which can mimic a genuine transit. ExoScan combines a data-driven classification ensemble with explicit transit-physics validation across four stages: (1) raw TESS light curves are downloaded and detrended to remove instrumental systematics; (2) the cleaned signal is phase-folded and screened by a dual-model ensemble classifying each candidate into transit, eclipsing binary, blend, or noise; (3) a Box Least Squares (BLS) physics validator independently fits orbital period, duration, and depth, and cross-checks dip symmetry and depth-radius consistency; (4) Monte Carlo Dropout produces a calibrated confidence score with explicit uncertainty bounds, alongside a signal-to-noise significance level.

2. Methodology

2.1 Data Sources

- Primary dataset: raw TESS light curves downloaded via the lightkurve API from the MAST archive (archive.stsci.edu/tess), drawn from a single high-cadence observation sector containing on the order of 20,000–30,000 individual stellar targets.
- Curated training labels: the BAH-provided curated dataset of known exoplanets, false positives, and eclipsing binaries, used to supervise the classification ensemble.
- Reference/validation set: NASA Kepler Cumulative KOI table, used as an independent source of confirmed planets with published orbital parameters for validating BLS parameter-estimation accuracy.

2.2 Detrending

Raw TESS flux exhibits low-frequency, ramp-like systematics from spacecraft thermal drift and detector response — not astrophysical in origin. ExoScan applies Savitzky-Golay filtering (lightkurve's flatten(), window length 401 cadences) to remove these trends while preserving short-duration transit dip shape, followed by 5-sigma outlier clipping for cosmic-ray hits. Savitzky-Golay was chosen over global polynomial detrending because its local sliding-window fit adapts to slow trend variation without flattening the transit signal itself.

2.3 Periodicity Search and Phase Folding

Each detrended light curve is searched for periodic dips using Box Least Squares (BLS) (Kovács, Zucker & Mazeh, 2002) via Astropy's BoxLeastSquares class. BLS was chosen over Lomb-Scargle because its box-shaped kernel matches transit dip geometry — a sharp, flat-bottomed flux reduction — rather than assuming sinusoidal periodicity. The period at maximum BLS power is used to phase-fold the light curve, amplifying signal-to-noise before classification and parameter fitting.

2.4 Classification Ensemble

Each phase-folded candidate is classified into transit, eclipsing binary, blend, or noise using a soft-voted ensemble:

- XGBoost on engineered features: depth, duration, period, impact-parameter proxy, SNR, and dip symmetry score.
- A 1D CNN on the raw phase-folded flux sequence, learning dip-shape patterns not fully captured by hand-engineered features.

Class imbalance is addressed via SMOTE on the training set only; the two models' class probabilities are averaged for the final prediction. The dual-model design combines XGBoost's interpretable physical statistics with the CNN's ability to recognize raw-shape patterns, reducing the risk of either model's blind spots dominating the decision.

2.5 Physics-Based Validation

A classification alone is insufficient evidence of a genuine transit. ExoScan applies four physical consistency checks to every candidate flagged as a transit or eclipsing binary:

- Periodicity strength: BLS power at the best-fit period must exceed a minimum significance threshold.

- Dip symmetry: a genuine transit produces a symmetric U-shaped dip; the folded dip's two halves are compared directly.
- U-shape vs. V-shape discrimination: eclipsing binaries typically produce a sharper V-shaped dip than a planetary transit's flatter U-shape, per the BAH mentor briefing; quantified via a flatness score.
- Depth-radius consistency: fractional depth implies a planet-to-star radius ratio, checked against an upper physical bound (~20 Earth radii) using catalog stellar radius, flagging implausibly large "planets."

A candidate is confirmed only if both the ensemble classifier and physics validator agree — this two-stage "AI + physics gate" is the primary mechanism reducing false positives relative to a classifier alone.

2.6 Parameter Estimation

For every confirmed transit, ExoScan reports orbital period (days), transit duration (hours), and transit depth (ppm) directly from the BLS fit at the best-fit period. These are validated against published parameters of known confirmed exoplanets in the Kepler KOI reference set using mean absolute percentage error (MAPE), reported in Section 3.2.

2.7 Confidence and Uncertainty Quantification

ExoScan reports calibrated confidence, not raw model probability. The XGBoost output is recalibrated via isotonic regression (CalibratedClassifierCV) so a reported 80% confidence corresponds to 80% empirical accuracy. The CNN runs 50 stochastic forward passes with dropout active at inference (Monte Carlo Dropout); the mean and standard deviation across passes yield confidence with an explicit uncertainty bound. MC Dropout was chosen over deep ensembles for compute efficiency — a single trained network approximates a predictive distribution. Each output reports confidence with ± uncertainty, alongside BLS-derived SNR as an independent significance measure.

2.8 Explainability

SHAP values are computed for the XGBoost component, providing per-prediction feature attribution so flagged candidates can be manually reviewed with a transparent rationale rather than an unexplained model output.

3. Results

Binary transit detection (XGBoost + 1D CNN on Kepler KOI validation set, n=1,053); 4-class TESS signal classification (Transit/EB/Blend/Noise, n=800 balanced).

3.1 Detection and Classification Performance

Class	Precision	Recall	F1-Score
Transit (Planet Candidate)	0.43	0.58	0.50
Eclipsing Binary	0.97	1.00	0.99
Blend	0.50	0.58	0.54
Noise / No Signal	0.40	0.18	0.25
Macro Average	**0.58**	**0.59**	**0.57**

Binary transit detection (Kepler validation, n=1,053): XGBoost F1=0.892, AUC=0.971, Precision=0.864, Recall=0.922; Ensemble F1=0.891, AUC=0.966. Physics gate independently rejects 55.5% of Kepler FALSE POSITIVE targets while approving 99.1% of CONFIRMED planets (n=7,316 evaluated, Cell [70]).

3.2 Parameter Estimation Accuracy

Estimated orbital period, transit duration, and transit depth are validated against published values for known confirmed exoplanets in the Kepler KOI reference set, using mean absolute percentage error (MAPE):

Parameter	Mean Absolute Percentage Error (MAPE)
Orbital Period	**4.44%**
Transit Duration	**9.37%**
Transit Depth	**7.97%**

3.3 Confidence Calibration

Reliability diagram analysis confirms the isotonic-calibrated XGBoost output tracks the perfect-calibration diagonal within 8.0 percentage points across confidence deciles. Monte Carlo Dropout uncertainty (standard deviation across 50 stochastic passes) is inversely correlated with classification accuracy, confirming that high-uncertainty predictions genuinely correspond to harder, less reliable cases rather than noise. The XGBoost model achieved a confidence of

0.9950 for a correctly-identified CONFIRMED planet (sample index 10, validation set), demonstrating the practical reliability of the calibrated output.

4. Assumptions and Limitations

- Single dominant transiting body per light curve is assumed; multi-planet disentangling is future work.
- Eclipsing-binary/blend examples come from the BAH curated dataset where available; limited representation is flagged in per-class reporting (3.1), not masked by aggregate accuracy.
- Depth-radius check assumes near-central transit; grazing transits may be conservatively flagged — a deliberate precision-over-recall choice given the cost of false confirmation.
- Stellar radius values are drawn from catalog entries (TIC / Kepler stellar parameters) and inherit their measurement uncertainty.
- Validated on a single TESS sector; cross-sector generalization is a priority for the Grand Finale phase.

5. Tools and Libraries

Tool / Library	Purpose
Lightkurve / Astropy	TESS download (MAST), detrending, phase folding, BLS fitting
XGBoost / PyTorch / Scikit-learn	Classifier ensemble, SMOTE, isotonic calibration, evaluation metrics
SHAP / NumPy / Pandas / Matplotlib	Feature attribution, numerical computation, visualization

6. References

[1] Shallue & Vanderburg (2018). Identifying Exoplanets with Deep Learning. AJ, 155(2), 94. — [2] Kovács, Zucker & Mazeh (2002). A box-fitting algorithm in the search for periodic transits. A&A, 391(1), 369–377. — [3] Ricker et al. (2015). Transiting Exoplanet Survey Satellite (TESS). JATIS, 1(1), 014003. — [4] Lundberg & Lee (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS 30. — [5] NASA Exoplanet Archive, Caltech. https://exoplanetarchive.ipac.caltech.edu